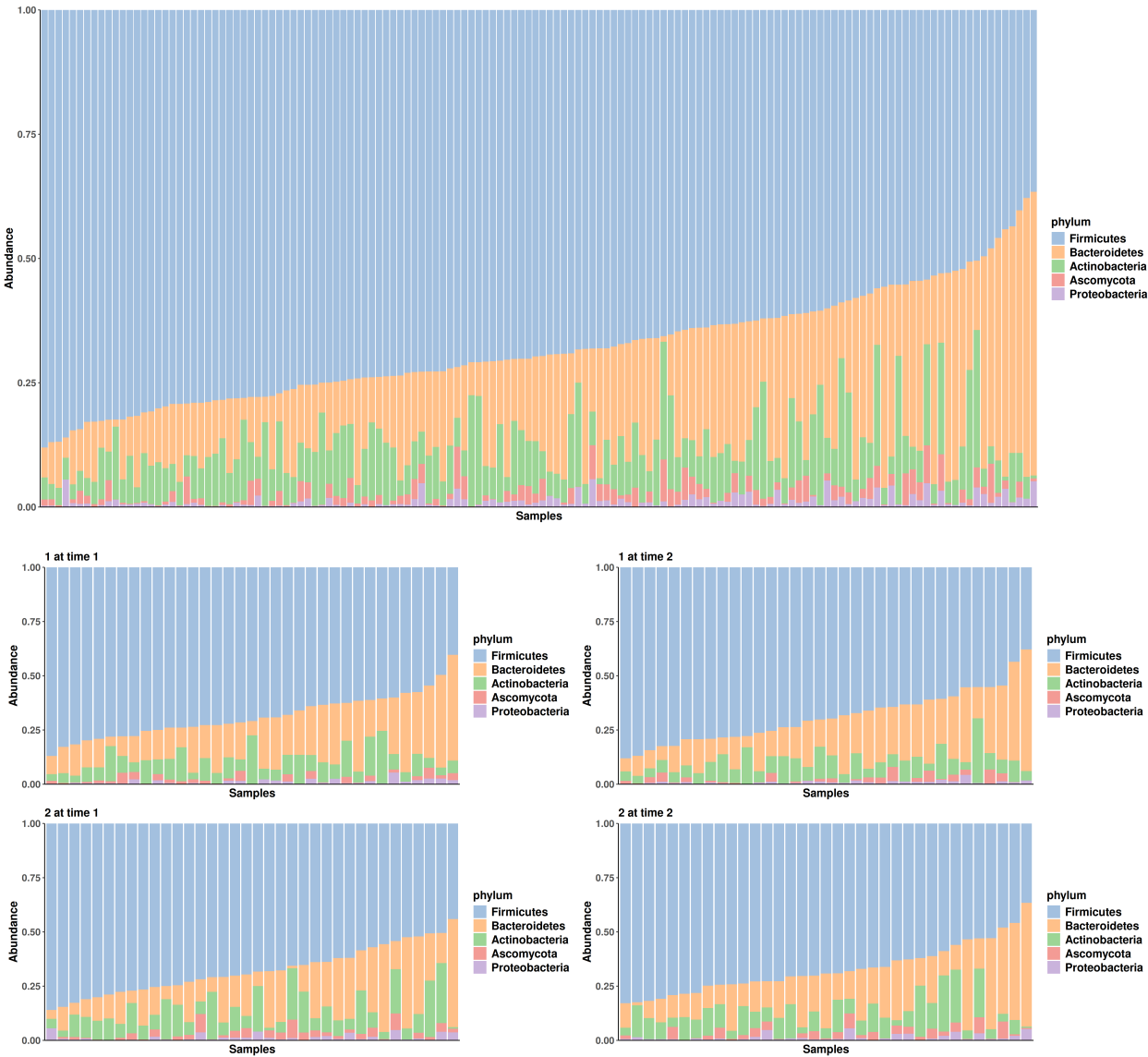


Beta diversity

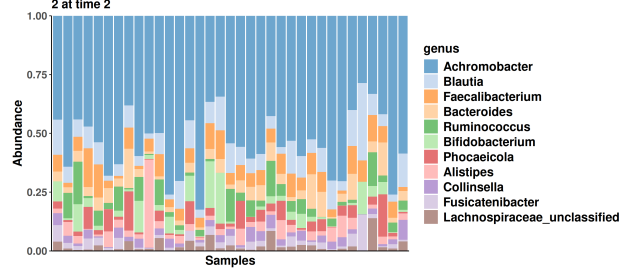
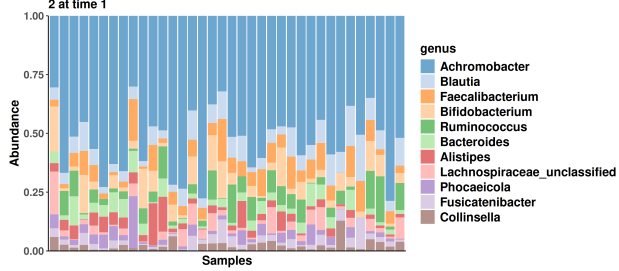
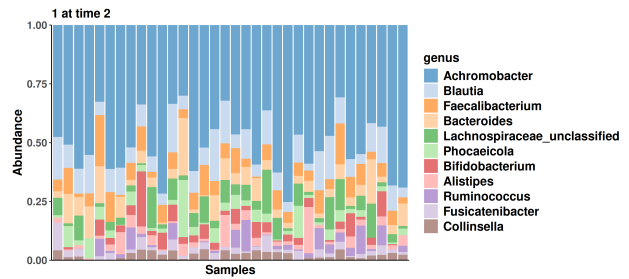
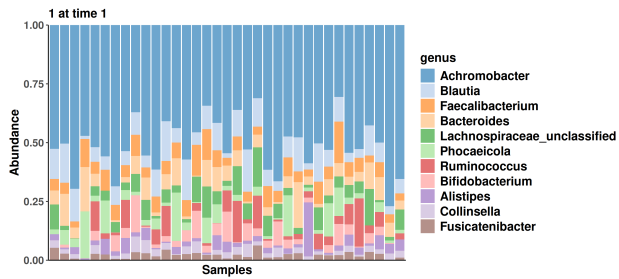
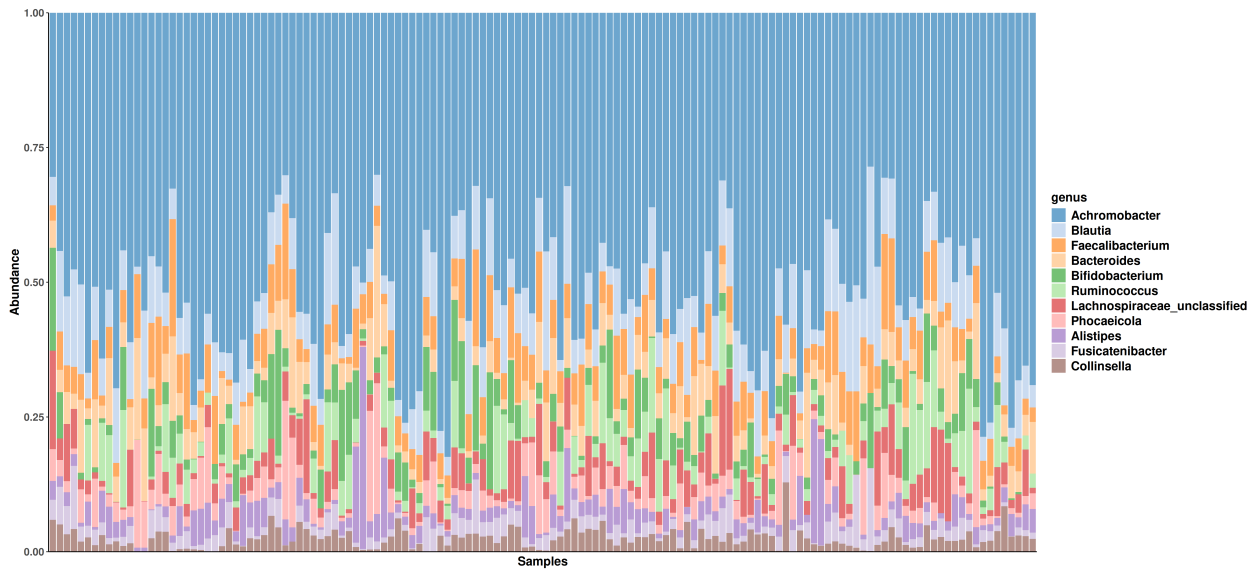
Muluh

1 Community composition

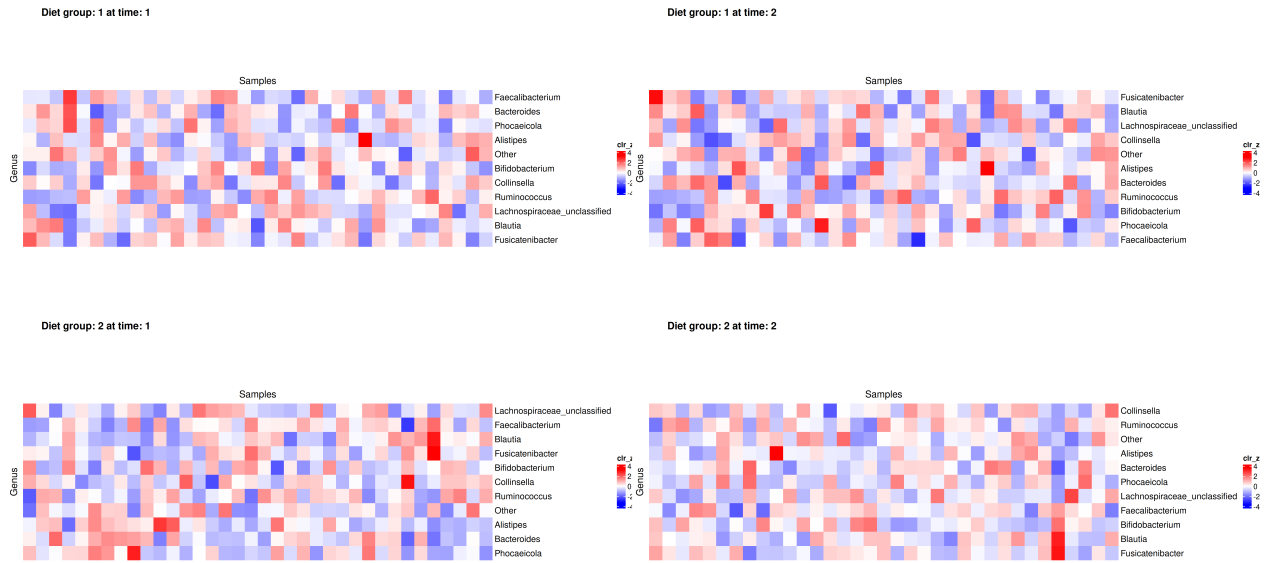
1.1 Phylum



1.2 Genus



1.3 Heatmap



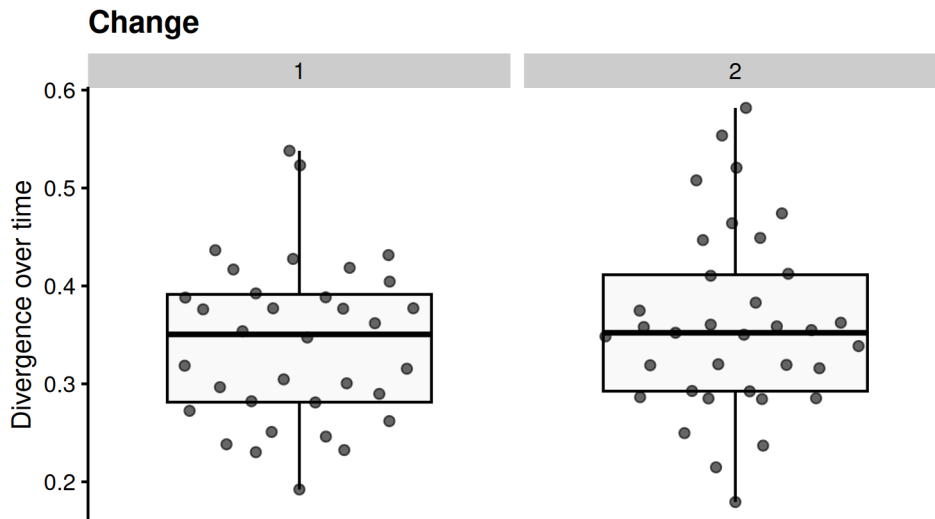
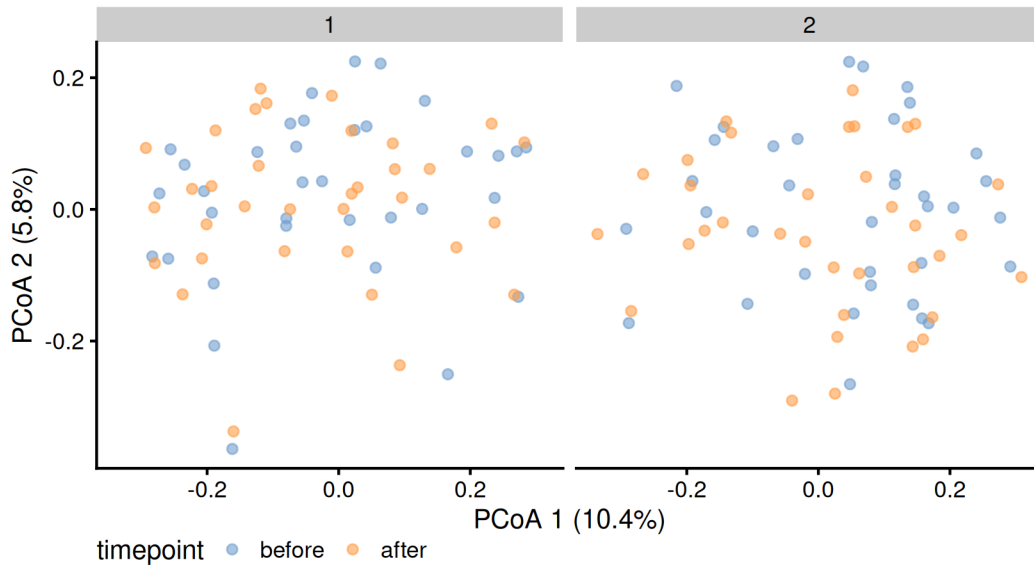
2 Unsupervised analysis

2.1 Community dissimilarity

This document presents the results of a microbiota project, focusing on beta diversity analysis. Specifically, in this section, we employed Principal Coordinates Analysis (PCoA) to assess community dissimilarity among microbiota samples.

- **Taxonomic level:** species
- **Dissimilarity index:** bray
- **Group variable:** diet
- **Multiple testing correction:** fdr
- **Statistical test:** Wilcoxon

2.2 PCoA plots



2.3 Table of the top findings

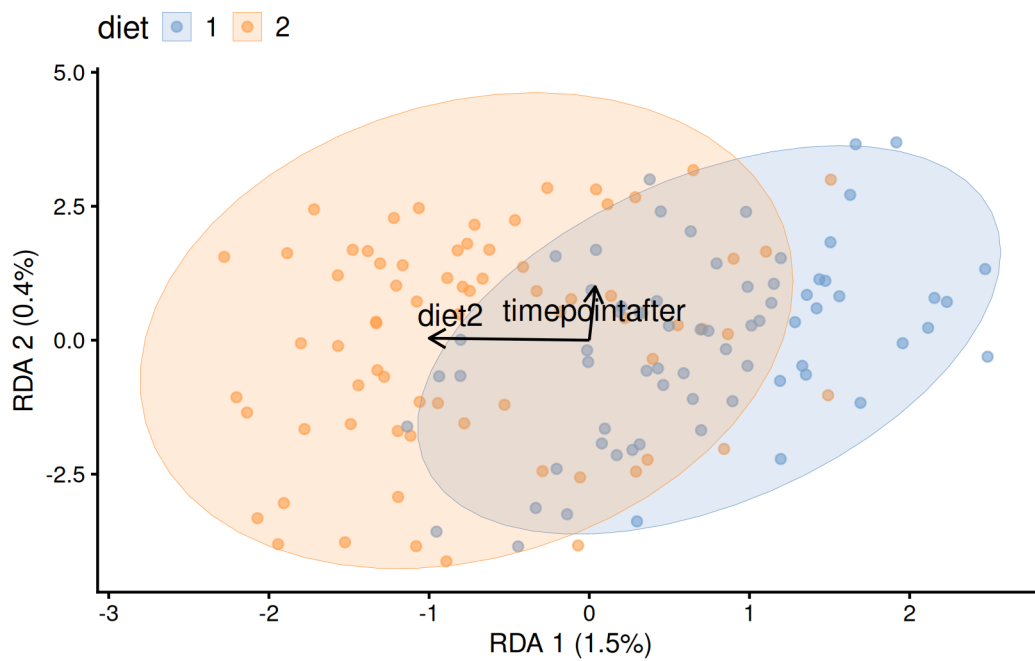
2.3.0.1 Paired treatment between timepoint(1 and 2)

- Multiple testing correction: **paired fdr**
- Statistical test: **Wilcoxon**

Table 1: Table of divergence paired between timepoint(1 and 2)

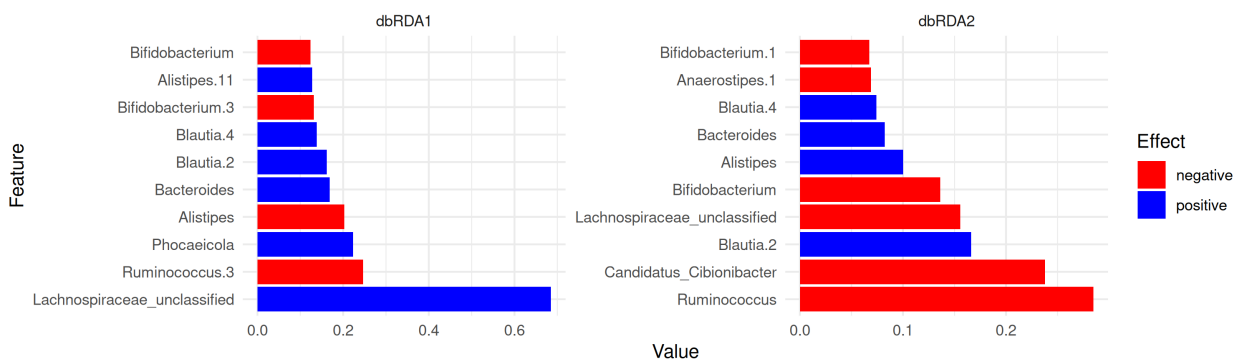
Meal	mean	p.value	p.adj
1	0.3426224	4e-07	4e-07
2	0.3612592	3e-07	4e-07

2.4 Significant variables



2.5 Significant taxa

- RDA



- MDS

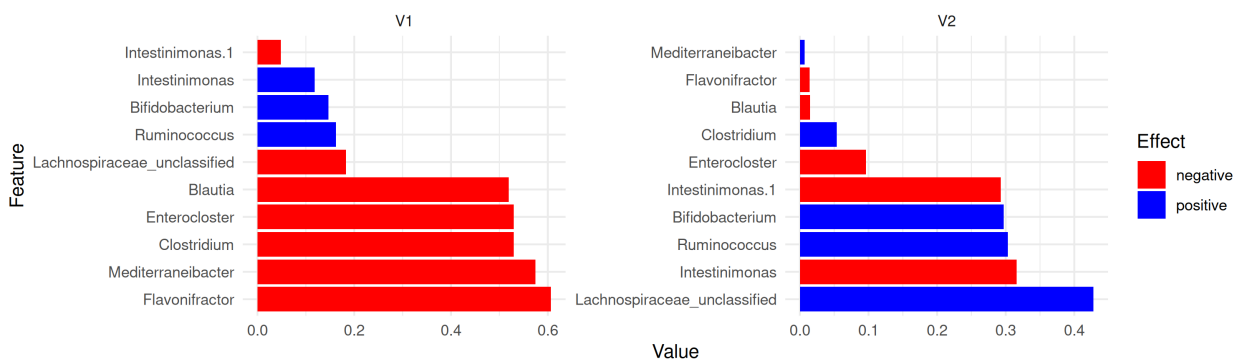


Table 2: PERMANOVA Results - before treatment

Group1	Group2	P_value	Homogeneity	p.adj
1	2	0.002	0.848	0.002

Table 3: PERMANOVA Results - after treatment

Group1	Group2	P_value	Homogeneity	p.adj
1	2	0.001	0.857	0.001

3 Supervised analysis

3.1 Permanova analysis

- Dissimilarity index: **bray**
- Group variable: **diet**

Interpretation

- Homogeneity < 0.05 : The assumption of homogeneity of variance is violated (i.e., the variances are significantly different across groups).
- Homogeneity > 0.05 : The assumption of homogeneity of variance is not violated (i.e., the variances are similar across groups).

3.1.1 Across-group

3.1.1.1 After treatment

Table 4: PERMANOVA Results - Paired between timepoint (1 and 2)

Diet	P_value	Homogeneity	p.adj
1	1.000	0.595	1
2	0.871	0.822	1

3.1.2 Within-group

4 Alternate analysis

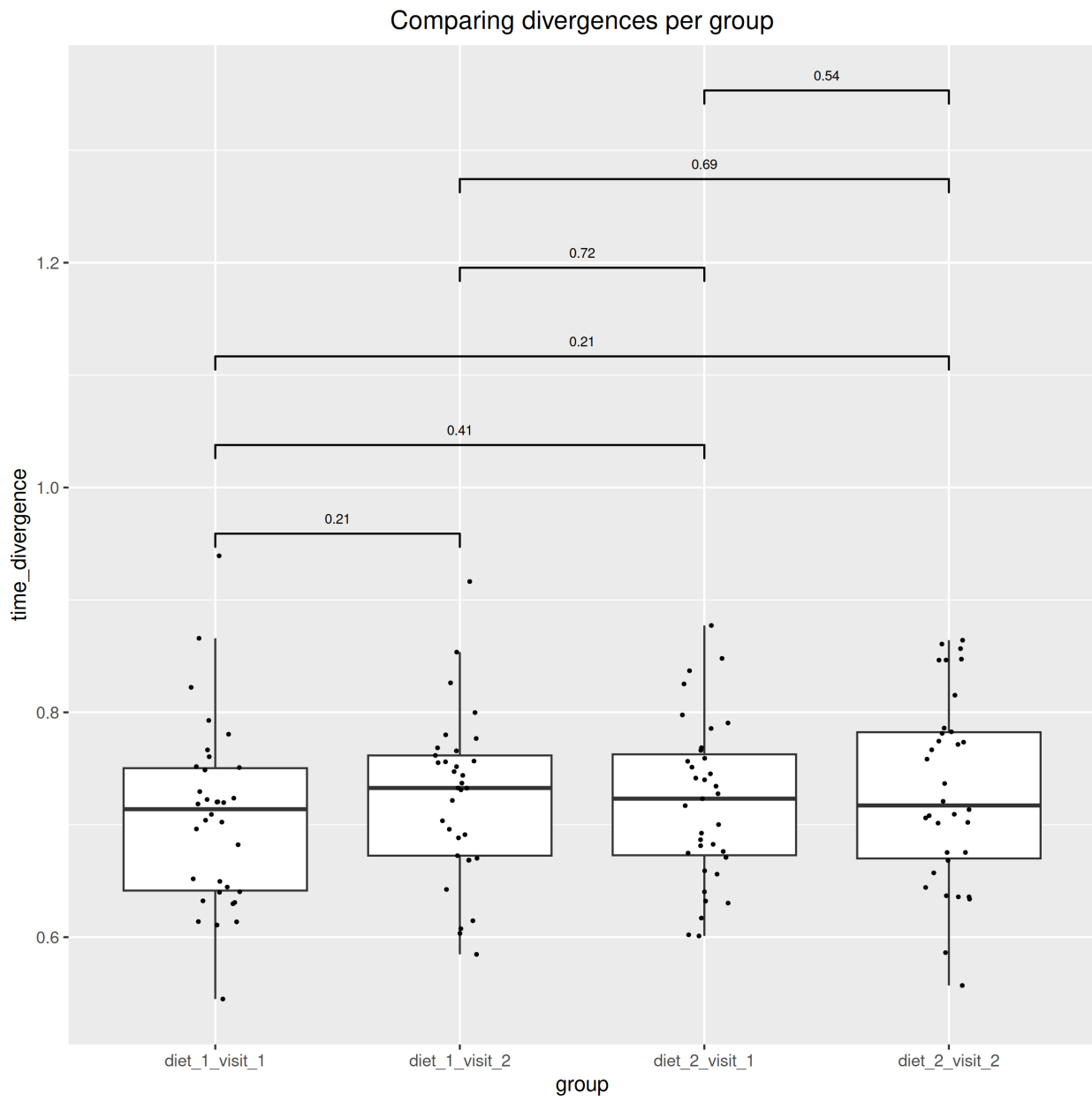


Table 5: PERMANOVA

	before	after	change-1	change-2	change-3	change-4
betadisp_pval	0.876	0.876	NA	NA	NA	NA
permanova_pval	0.332	0.332	1	0.978	0.992	1

4.1 Data Summary

```

class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921

```

```
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(16): sample id ... time_difference time_divergence
reducedDimNames(1): PCoA_BC
mainExpName: NULL
altExpNames(14): kingdom phylum ... KO metacyc
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```